

3.4 PN Junction: Diode Models

3.4.1 Load-Line Analysis

3.4.2 Ideal Diode Model

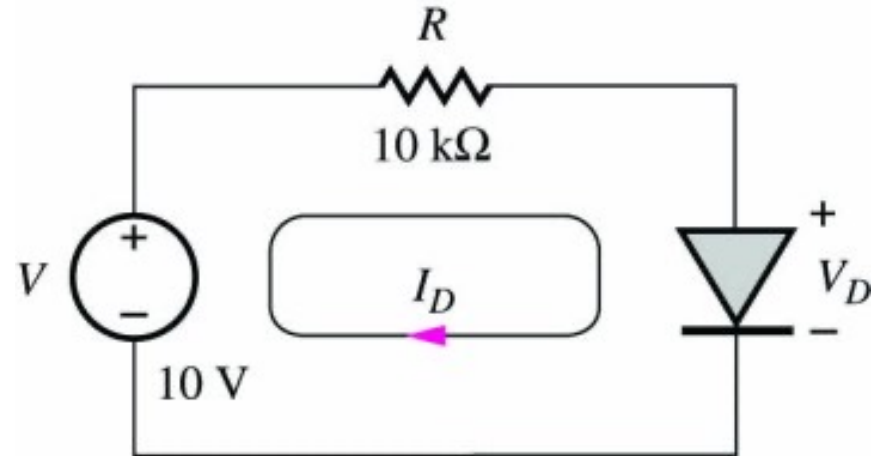
3.4.3 Constant Voltage Drop Model

3.4.4 Breakdown Model

Jaeger & Blalock, Chapter 3.10-3.12, page 96-112

Diode Circuit Analysis

- Establish **simple diode models** to facilitate (dc) analysis of circuits containing diodes
- Example: Series circuit of voltage source, resistor and diode
- Note: V and R may represent **Thévenin equivalent** of more complicated two-terminal network
- Goal: Find **quiescent operating point** or **Q-point** of diode



Jaeger, Blalock, Fig. 3.22

$$V = I_D R + V_D$$

What are the values for V_D and I_D ?

3.4.1 Load-Line Analysis

- Load-Line Analysis = **Graphical analysis** if diode I-V characteristic is only available in graphical form

Analysis approach:

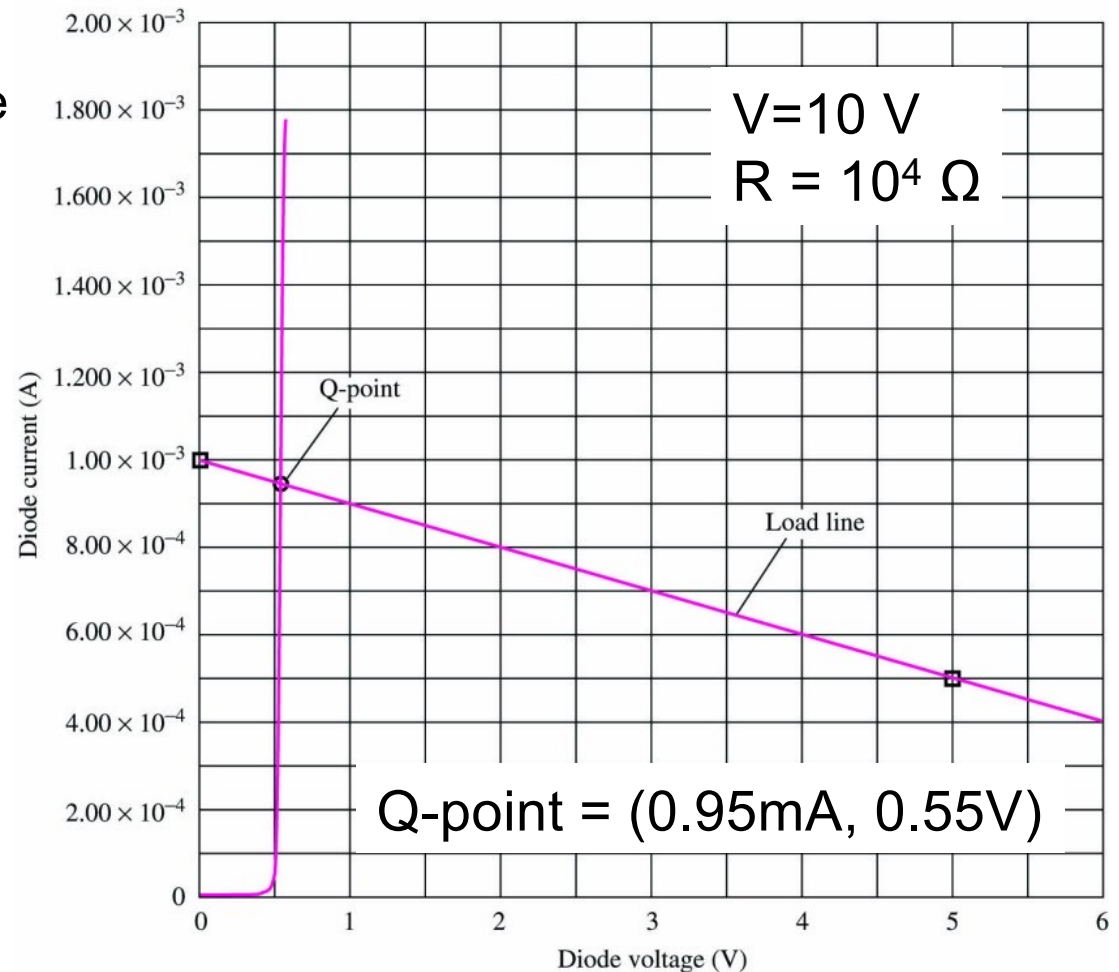
- Write load-line equation

$$V = I_D R + V_D$$

$$10 = I_D 10^4 + V_D$$

$$I_D = -10^{-4} V_D + 10^{-3}$$

- Plot load-line and diode I-V characteristic
- Q-point is defined by intersection of load line and diode I-V characteristic



Jaeger, Blalock, Fig. 3.23

Load-Line Analysis

– The Mathematical Approach –

- Goal: **Find intersection** of load line and diode I-V characteristic using mathematical software package

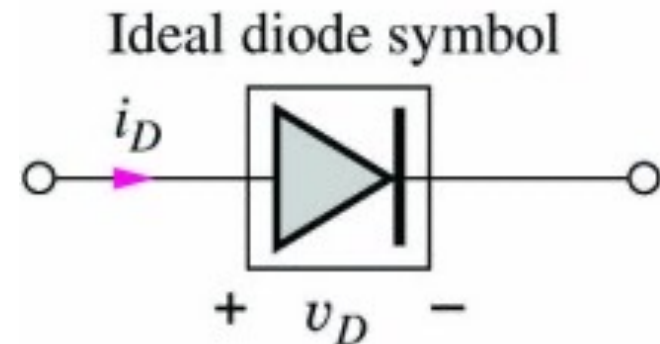
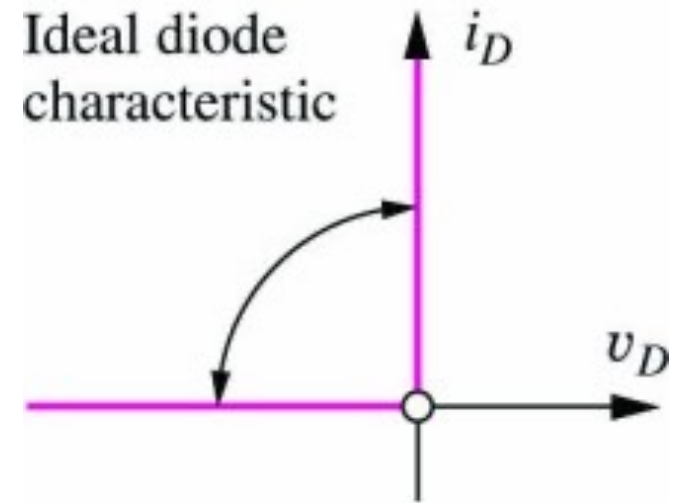
$$I_D = -\frac{1}{R} V_D + \frac{V}{R}$$

$$I_D = I_S \left[e^{qV_D/kT} - 1 \right]$$

- Given are $R = 10^4 \, \Omega$, $V = 10 \, \text{V}$, $I_S = 10^{-13} \, \text{A}$ and $kT/q = 0.026 \, \text{V}$, unknown are V_D and I_D
- The **Mathematica** function ***FindRoot*** yields $V_D = 0.597 \, \text{V}$ and $I_D = 0.940 \, \text{mA}$

3.4.2 Ideal-Diode Model

- **Approximation** of non-linear diode characteristic by **piece-wise linear model**
- **Ideal Diode Model =** Approximation with two straight-line segments
- If diode is **forward-biased**, voltage across diode is zero
 $v_D = 0$ for $i_D > 0$, i.e. **short circuit**
- If diode is **reverse-biased**, current through diode is zero
 $i_D = 0$ for $v_D < 0$, i.e. **open circuit**
- Diode is assumed to be either on or off

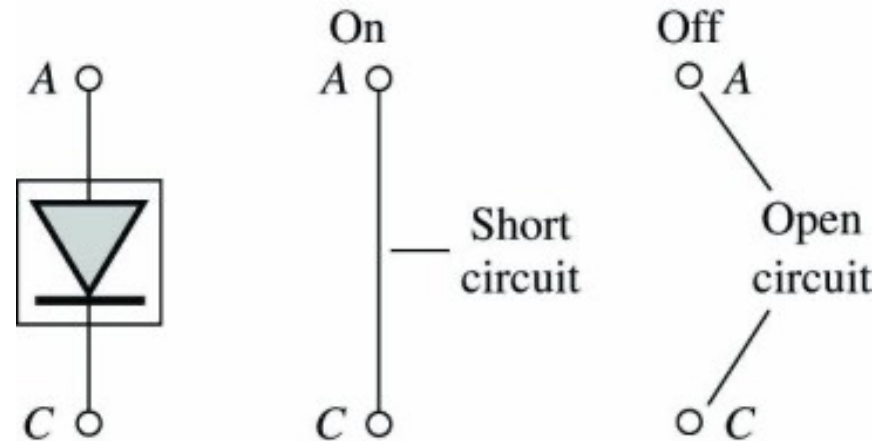


Jaeger, Blalock, Fig. 3.26

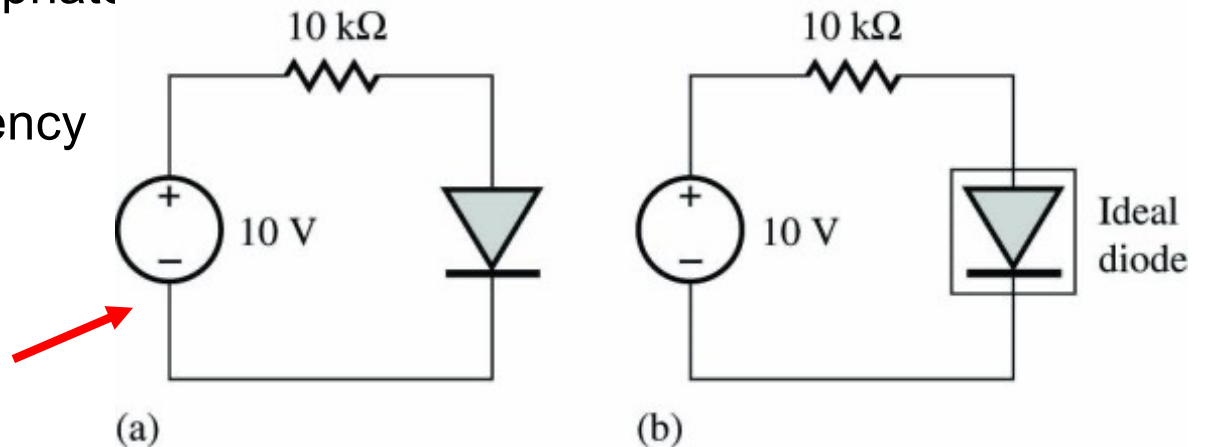
Ideal Diode Model

Analysis procedure:

- Select model for diode
- Identify diode anode (positive terminal) and cathode (negative terminal) and label V_D and I_D
- Make (educated) guess on diode region of operation based on circuit configuration
- Analyze circuit with appropriate diode model
- Check results for consistency with assumptions
- What about our simple example?

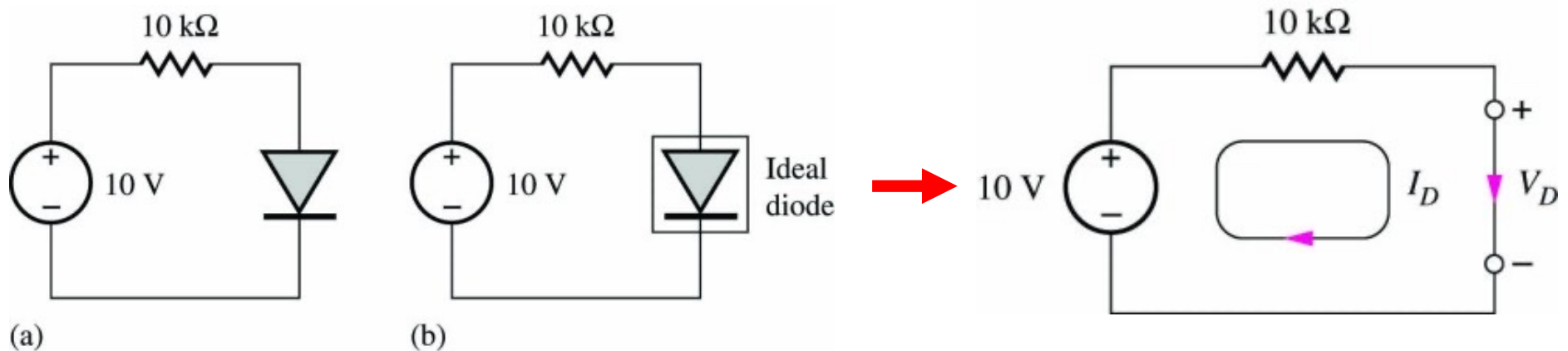


Jaeger, Blalock, Fig. 3.26



Jaeger, Blalock, Fig. 3.27

Example: Analysis with Ideal-Diode Model



Jaeger, Blalock, Fig. 3.27&28

- Voltage source tries to forward-bias diode, so we replace it by short-circuit (ON state)

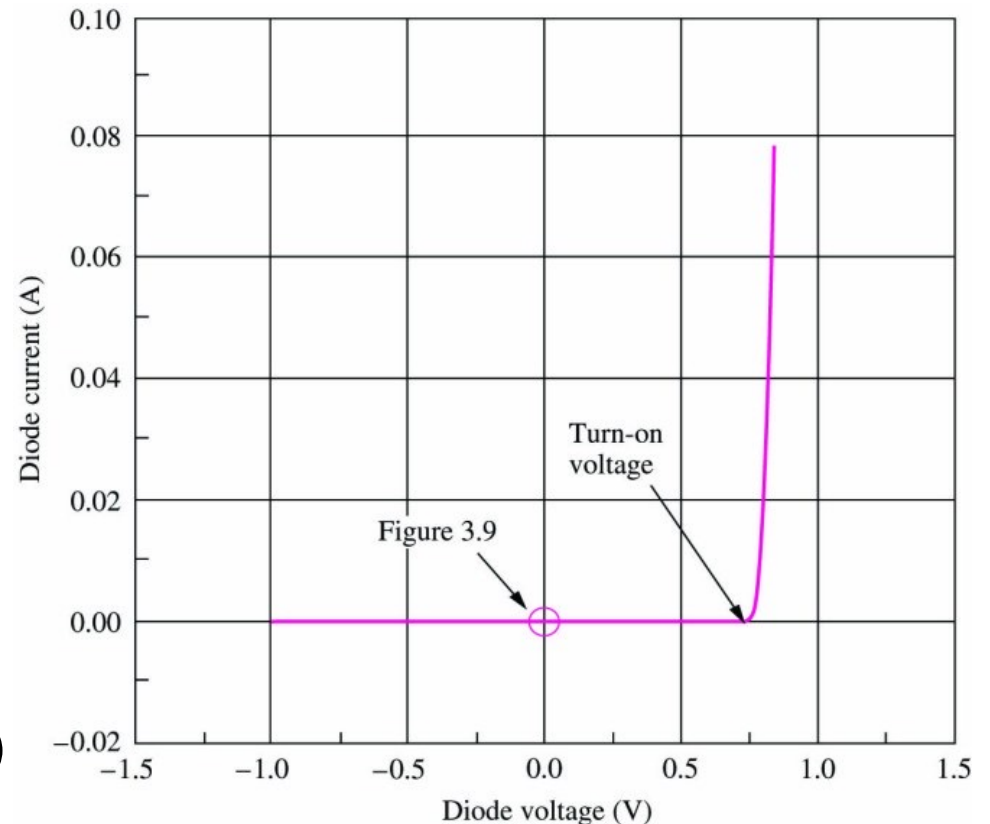
- I_D is now easily obtained:
$$I_D = \frac{V}{R} = 1 \text{ mA}$$

$$V_D = 0 \text{ V}$$

- The current (positive) is consistent with the assumption that the diode is ON

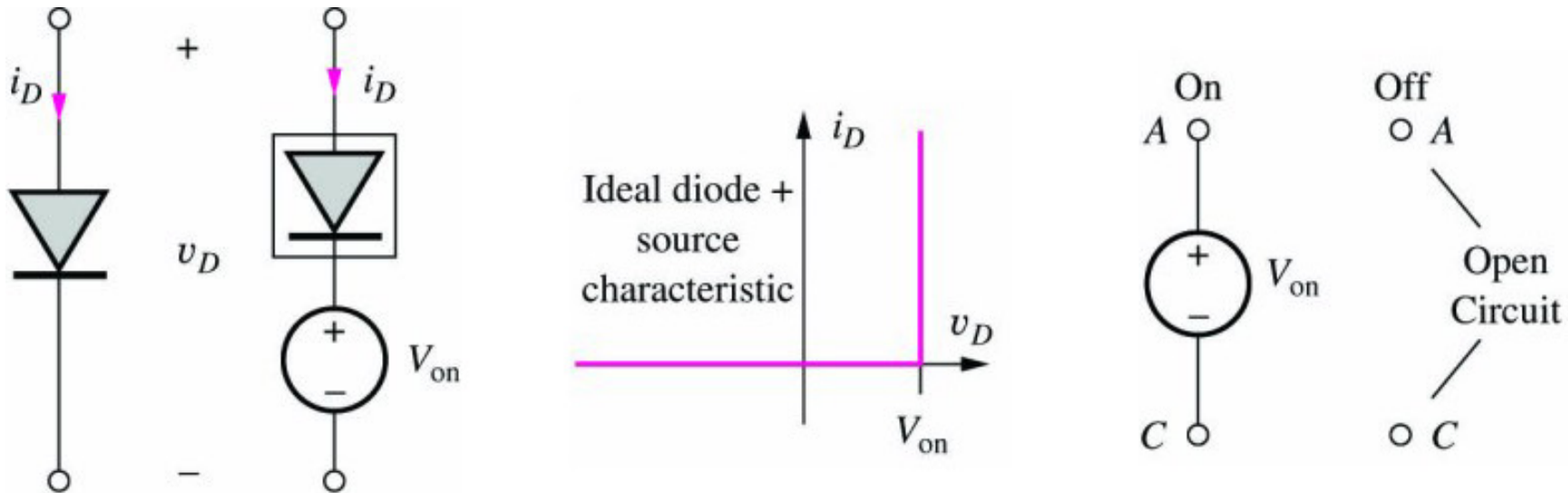
3.4.3 Constant Voltage Drop Model

- Piecewise linear model accounting for “constant” voltage drop across forward-biased diode (associated with **turn-on voltage** of diode)
- Turn-on voltage = voltage required to obtain “significant conduction”; typically 0.5-0.7 V
- Constant voltage drop (CVD) model: simulation of diode by **series connection of ideal diode and voltage source**



Jaeger, Blalock, Fig. 3.8

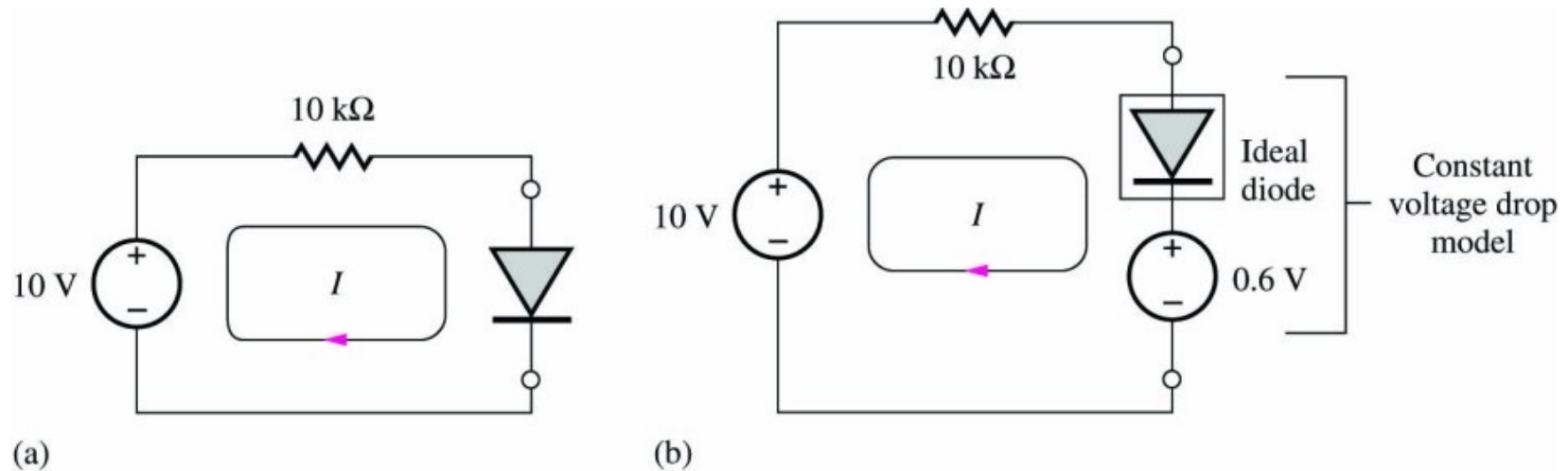
Constant Voltage Drop Model



Jaeger, Blalock, Fig. 3.31

- If diode is forward-biased, voltage across diode is turn-on voltage **$v_D = V_{on}$ for $i_D > 0$** , i.e. **constant voltage source**
- If diode is reverse-biased, current through diode is zero **$i_D = 0$ for $v_D < V_{on}$** , i.e. **open circuit**
- We choose **$V_{on} = 0.6 \text{ V}$**

Example: Analysis with CVD Model

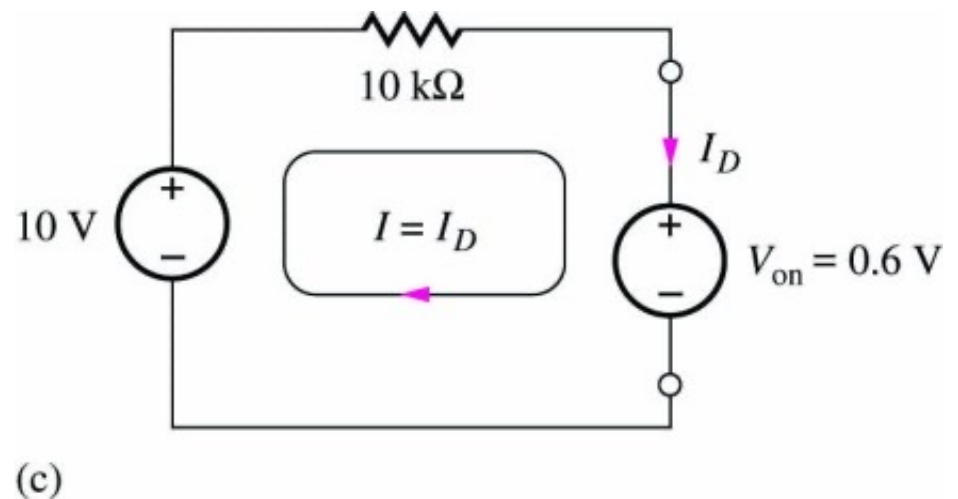


Jaeger, Blalock, Fig. 3.32

- Voltage source tries to forward-bias diode, so we replace it by voltage source (ON state)
- I_D is again easily obtained:

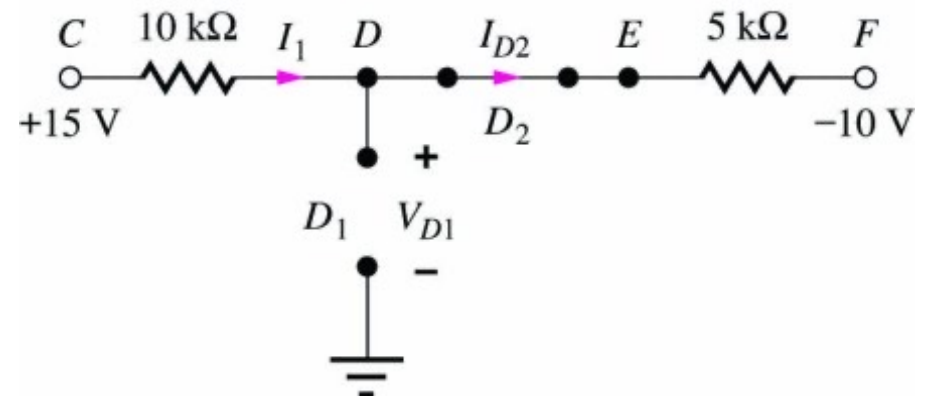
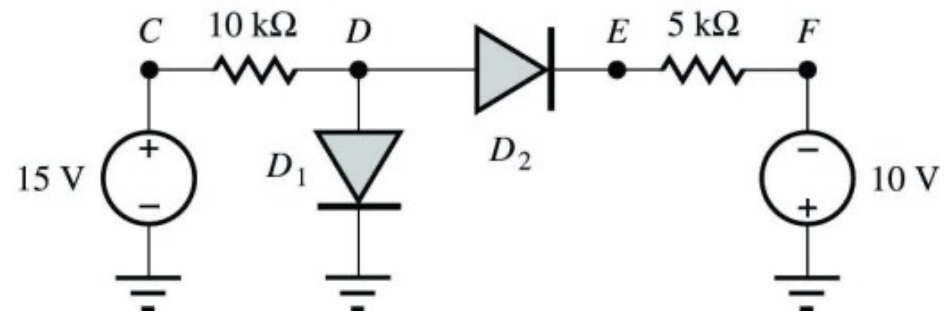
$$I_D = \frac{V - V_{on}}{R} = 0.94 \text{ mA}$$

$$V_D = 0.6 \text{ V}$$



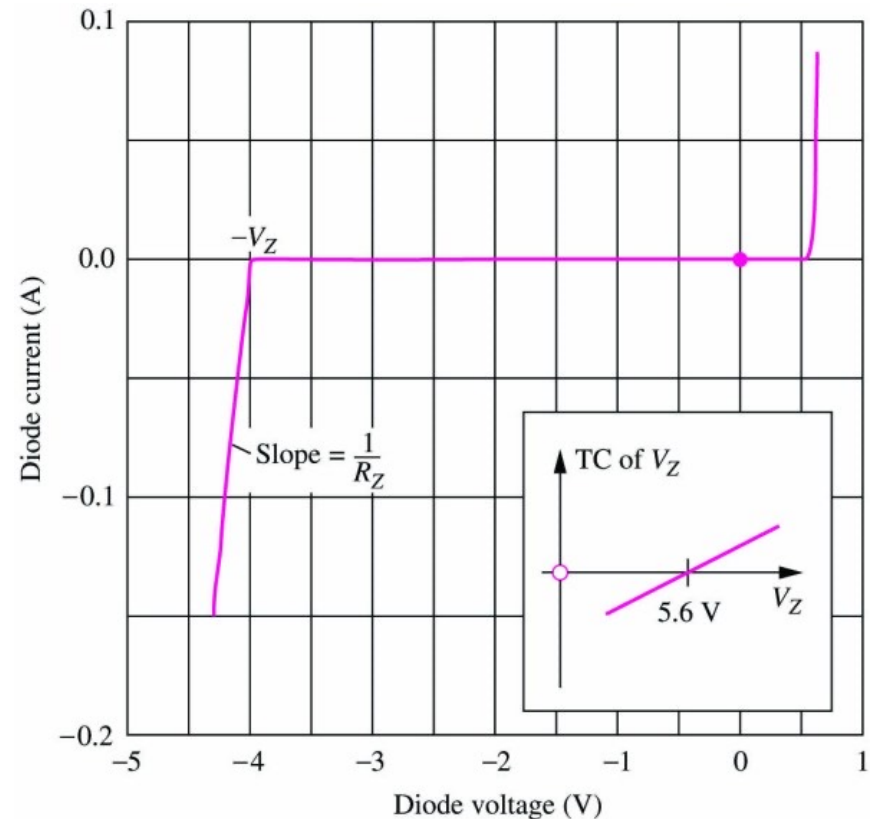
Multi-Diode Circuits

- Ideal diode and CVD model enable hand-analysis of more complex diode circuits having two or more diodes
- Alternative: **SPICE analysis** (see Chapter 3.6)
- Example problem: Find Q-points of both diodes in two-diode circuit!
- Approach: Try all (four) possible diode states (one after the other) and check whether results are consistent with assumptions
- Result (using ideal diode model):
D1 off: (0 mA; -1.67 V)
D2 on: (1.67 mA; 0 V)



3.4.4 Breakdown Model

- **Zener diodes** are
 - diodes designed to be **operated in reverse breakdown** (with proper current limiter to protect diode)
 - used as **voltage regulators** or **voltage reference** in electronic circuits
- In the breakdown regime, the diode can be modeled by **voltage source V_Z in series with resistor R_Z**
- Voltage across diode in reverse breakdown is almost constant and independent of current



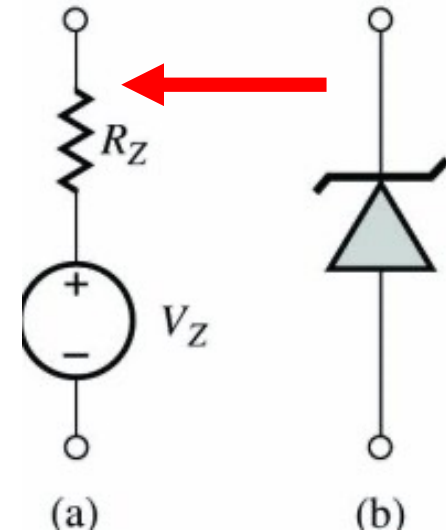
Jaeger, Blalock, Fig. 3.14

Breakdown Model for Zener Diode

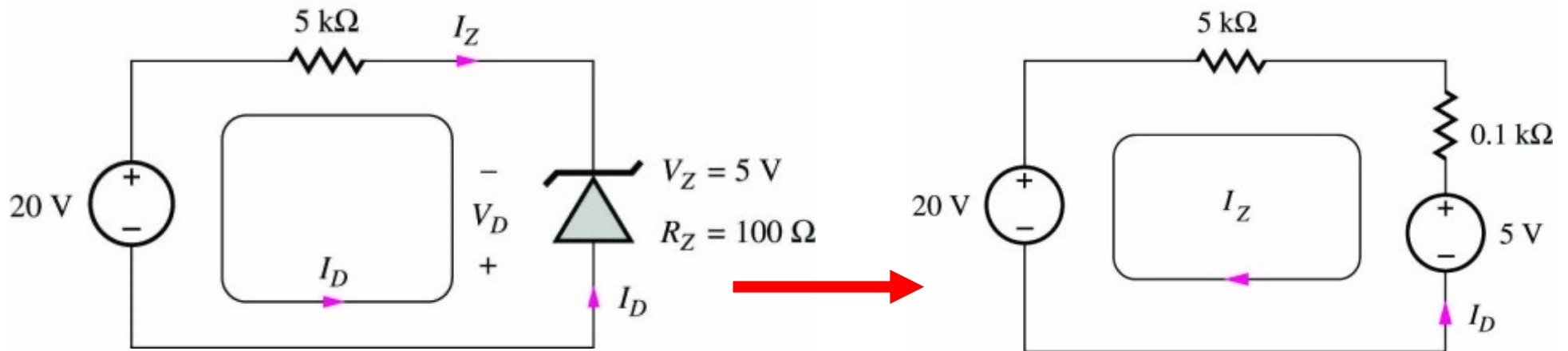
- Example: Series circuit of voltage source, resistor and Zener diode
- Zener diode replaced by voltage source V_Z in series with resistor R_Z
- Current I_Z becomes:

$$I_Z = \frac{20 - 5 \text{ V}}{5100 \Omega} = 2.94 \text{ mA}$$

- Note: Diode has actually 3 piecewise linear regimes



Jaeger, Blalock, Figs. 3.16, 3.36 & 3.38



Voltage Regulator with Zener Diode

- In a voltage regulator, the Zener diode maintains a constant voltage ($\approx V_Z$) across the load resistor R_L (even if R_L changes)
- Circuit analysis:

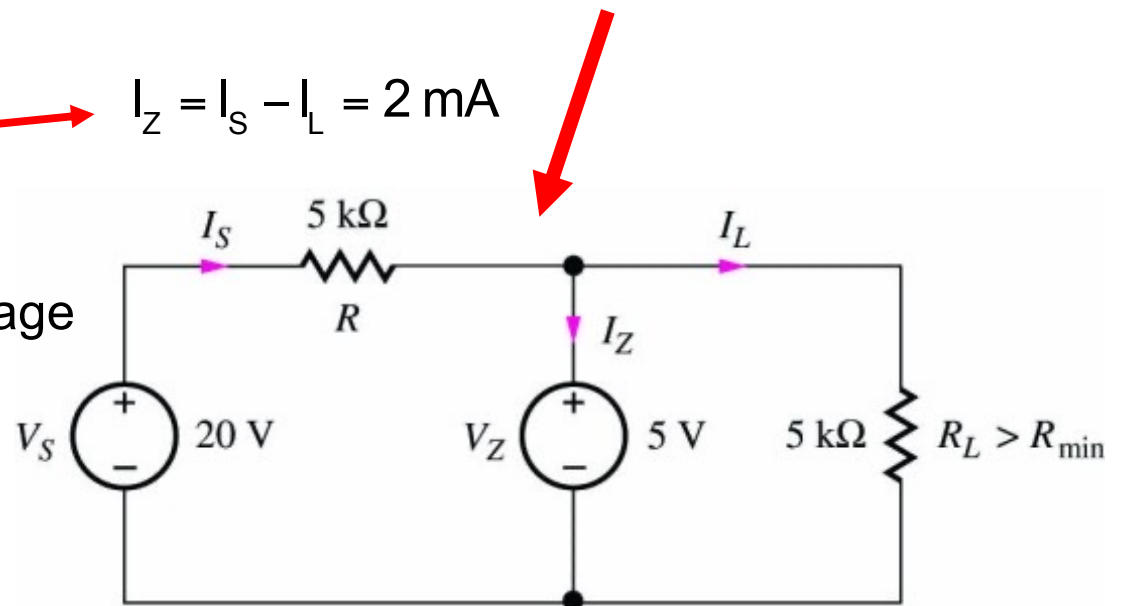
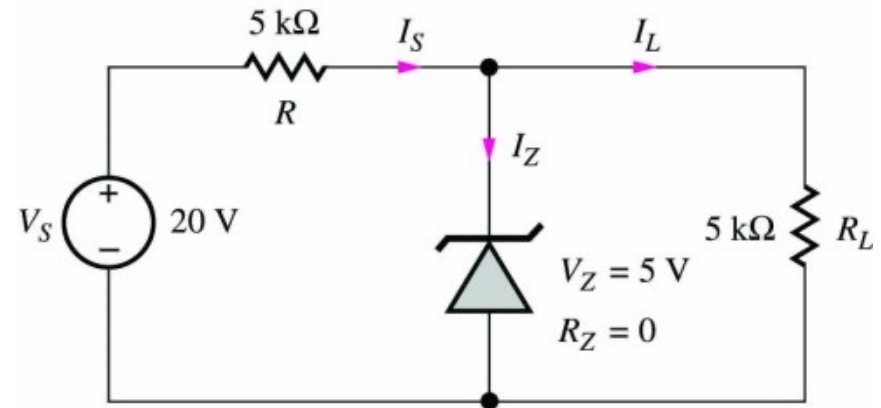
$$I_S = I_Z + I_L$$

$$I_S = \frac{V_S - V_Z}{R} = \frac{20 - 5 \text{ V}}{5000 \Omega} = 3 \text{ mA}$$

$$I_L = \frac{V_Z}{R_L} = \frac{5 \text{ V}}{5000 \Omega} = 1 \text{ mA}$$

- Zener diode maintains voltage regulation as long as I_Z is positive, yielding

$$R_L > \frac{R}{\frac{V_S}{V_Z} - 1} = 1.67 \text{ k}\Omega$$



Jaeger, Blalock, Figs. 3.39 & 3.40